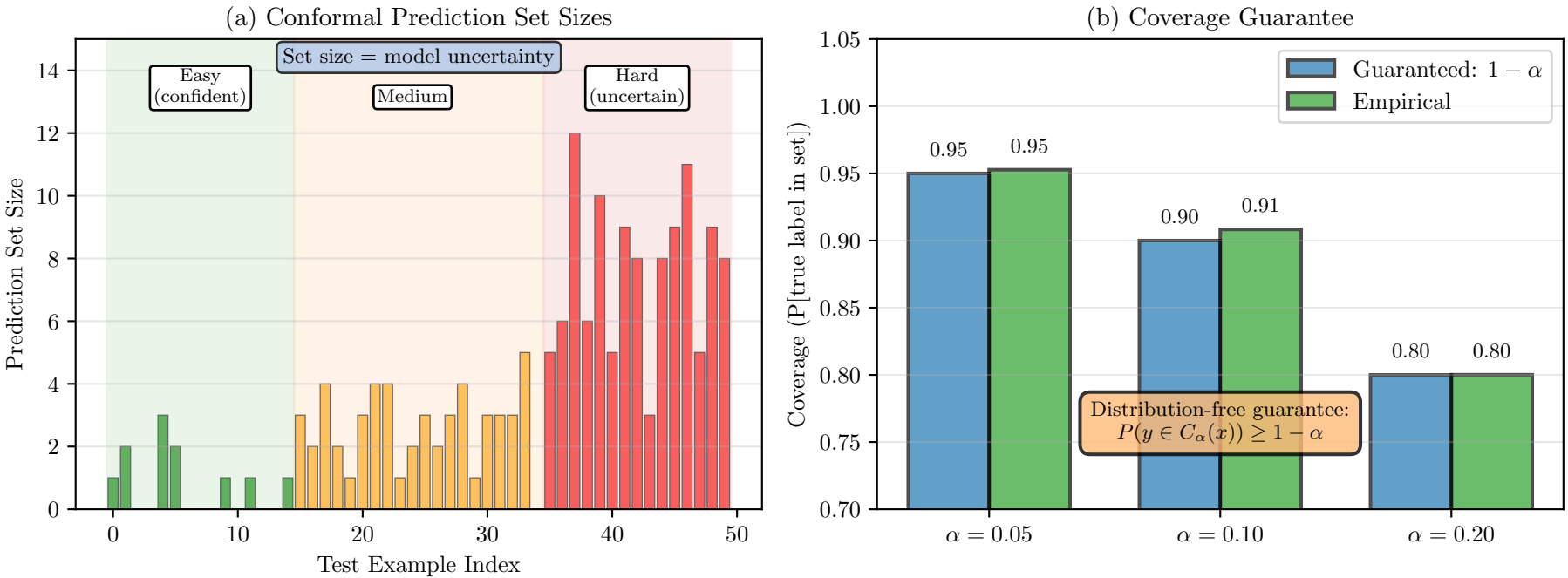
Conformal Prediction: Rigorous Coverage Guarantees
(a) Conformal Prediction Set Sizes
Set size = model uncertainty
Easy (confident)
Medium
Hard (uncertain)
Prediction Set Size
Test Example Index
0
2
4
6
8
10
12
14
0
10
20
30
40
50
(b) Coverage Guarantee
Guaranteed: $1-\alpha$
Empirical
Coverage (P[true label in set])
0.95
0.95
0.90
0.91
0.80
0.80
Distribution-free guarantee:
$P(y \in C_\alpha(x)) \geq 1-\alpha$
$\alpha = 0.05$
$\alpha = 0.10$
$\alpha = 0.20$
0.70
0.75
0.80
0.85
0.90
0.95
1.00
1.05